BABU BANARASI DAS UNIVERSITY

2026-2027



SCHOOL OF COMPUTER APPLICATIONS

No SQL Project Report On

StudentManagementSystemUsing MongoDB

Submitted to:
Mr. HarendraSingh
Department:BCA (DS &AI)
Date: 25th September

Submitted by:
Kartikey Tiwari
Roll no: (1250258240)

ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher Mr. Harendra Singh for giving me the opportunity to work on this NoSQL project titled “Student Management System Using MongoDB”.

This project helped me understand the basic concepts of NoSQL databases and learn how MongoDB is used to create, retrieve, update, Sorting and delete data. I also learned how different query operators such as greater than, less than, equal to, AND, and OR are used in MongoDB.

I am thankful to my teacher for the guidance and support provided throughout the completion of this project.

INTRODUCTION

MongoDB is a popular NoSQLdatabase managementsystem that stores data in the form of documents. Unlike traditional relational databases, MongoDB does not require data to be stored in fixed rows and columns.

In this project, a Student Management System has been created using MongoDB. The database contains information about 50 students, including their roll number, name, age, marks, and city.

Different MongoDB operations have been performed on the student data, such as inserting, displaying, updating, deleting, sorting, counting, and searching records. Comparison operators such as $eq, $gt, $lt, $gte, and $lte have also been demonstrated.

OBJECTIVES

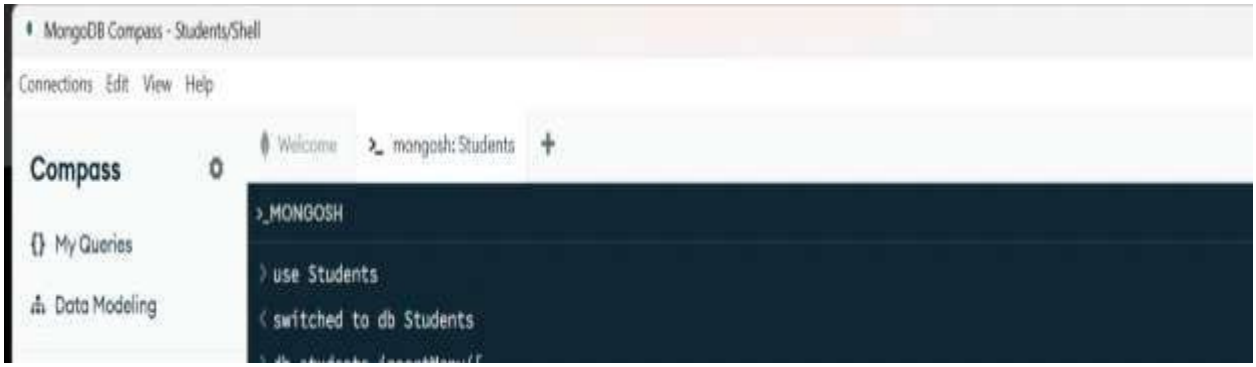
- To understand the concept of NoSQL databases.
- To learn the basic features of MongoDB.
- To create a database and collection.
- To insert 50 student records.
- To perform CRUD operations. • To use comparison operators.
- To perform sorting and filtering.
- To understand AND and OR operations.

1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Student database.

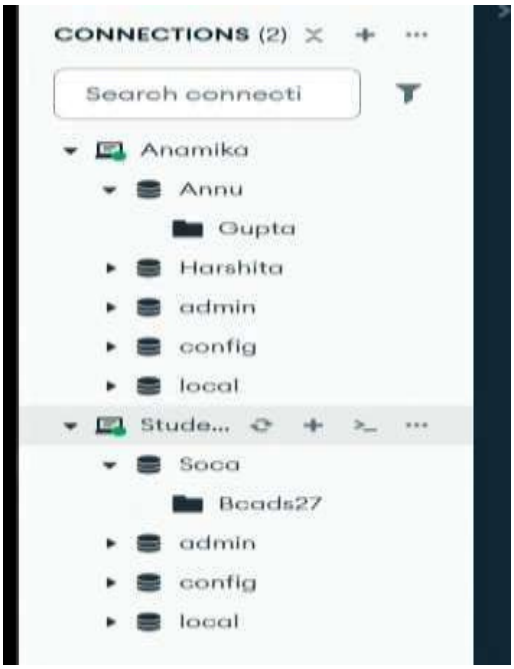


2. CREATE COLLECTION

MongoDB Query:

db. create Collection("students")

Explanation: Creates a collection named students.



3. INSERT 50 STUDENTS

MongoDB Query:

db. students. Insert Many([ ])

{roll:1, name:"Anamika", age:19,marks:88,city:"Lucknow"},

{roll:2, name:"Uday", age:18,marks:76,city:"Jaunpur"},

{roll:3, name:"Vaishnavi", age:19,marks:92,city:"Kanpur"},

{roll:4, name:"Annu", age:18,marks:69,city:"Ayodhya"},

{roll:5, name:"Vansh", age:19,marks:81,city:"Varanasi"},

{roll:6, name:"Amrita", age:20,marks:85,city:"Lucknow"},

{roll:7, name:"Naveen", age:19,marks:73,city:"Prayagraj"},

{roll:8, name:"Pratik", age:19,marks:90,city:"Lucknow"},

{roll:9, name:"Sangita", age:20,marks:78,city:"Gorakhpur"},

{roll:10, name:"Vivek", age:18,marks:65,city:"Kanpur"},

{roll:11, name:"Tarkeshwar", age:20,marks:71,city:"Jaunpur"},

{roll:12, name:"Shreya", age:19,marks:95,city:"Lucknow"},

{roll:13, name:"Akshita", age:18,marks:84,city:"Ayodhya"},

{roll:14, name:"Shruti", age:19,marks:87,city:"Prayagraj"},

{roll:15, name:"Kartik", age:20,marks:74,city:"Varanasi"},

{roll:16, name:"Sejal", age:18,marks:89,city:"Lucknow"},

{roll:17, name:"Astha", age:19,marks:82,city:"Kanpur"},

{roll:18, name:"Jai", age:18,marks:58,city:"Gonda"},

{roll:19, name:"Harshita", age:20,marks:91,city:"Lucknow"},

{roll:20, name:"Omkar", age:19,marks:63,city:"Bareilly"},

{roll:21, name:"Ayush", age:18,marks:77,city:"Lucknow"},

{roll:22, name:"Saumya", age:19,marks:86,city:"Jaunpur"},

{roll:23, name:"Vidya", age:20,marks:72,city:"Kanpur"},

{roll:24, name: "Adarsh", age:19,marks:80,city:"Ayodhya"},

{roll:25, name: "Anmol", age:18,marks:67,city:"Prayagraj"},

{roll:26, name: "Anushka", age:20,marks:94,city:"Lucknow"},

{roll:27, name: "Ananya", age:19,marks:88,city:"Varanasi"},

{roll:28, name: "Bhaskar", age:18,marks:61,city:"Gorakhpur"},

{roll:29, name: "Rohan", age:19,marks:75,city:"Kanpur"},

{roll:30, name: "Nishant", age:20,marks:79,city:"Lucknow"},

{roll:31, name: "Vineet", age:18,marks:68,city:"Jaunpur"},

{roll:32, name: "Anika", age:19,marks:90,city:"Lucknow"},

{roll:33, name: "Jai", age:20,marks:55,city:"Ayodhya"},

{roll:34, name: "Lakshya", age:18,marks:83,city:"Kanpur"},

{roll:35, name: "Krishna", age:19,marks:96,city:"Prayagraj"},

{roll:36, name: "Rumana", age:20,marks:70,city:"Lucknow"},

{roll:37, name: "Priyanshu", age:18,marks:81,city:"Varanasi"},

{roll:38, name: "Aditya", age:19,marks:66,city:"Gorakhpur"},

{roll:39, name: "Dhruv", age:20,marks:93,city:"Lucknow"},

{roll:40, name: "Nivedita", age:19,marks:85,city:"Kanpur"},

{roll:41, name: "Meenu", age:18,marks:74,city:"Jaunpur"},

{roll:42, name: "Akanksha", age:20,marks:89,city:"Lucknow"},

{roll:43, name: "Sarika",age:19,marks:62,city:"Ayodhya"},

{roll:44, name: "Anchal",age:18,marks:78,city:"Prayagraj"},

{roll:45, name: "Umra", age:20,marks:84,city:"Lucknow"},

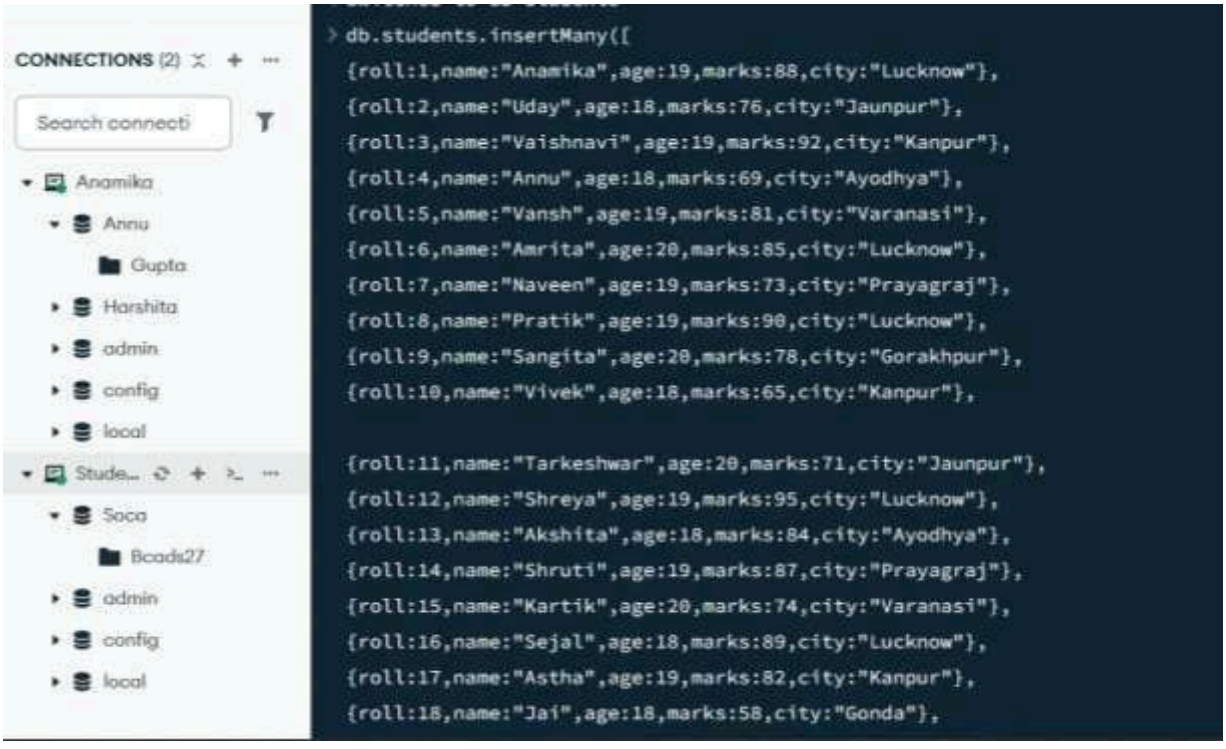
{roll:46, name: "Nashra", age:19,marks:73,city:"Kanpur"},

{roll:47, name: "Aman", age:18,marks:59,city:"Gonda"},

{roll:48, name: "Priya", age:20,marks:87,city:"Lucknow"},

{roll:49, name: "Rahul", age:19,marks:64,city:"Bareilly"},

{roll:50, name: "Neha", age:18,marks:91,city:"Varanasi"}



Explanation: Inserts multiple student documents into the collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

db. students. Find ()

Explanation: Retrieves documents from the students

collection.



5. EQUAL TO — $eq

MongoDB Query:

db.students.Find({marks: {$eq: 88}})

Explanation: Finds students whose marks are exactly 88.



6. GREATER THAN — $gt

MongoDBQuery:

db.students.find({marks: {$gt: 85}})

Explanation: Finds students whose marks are greater than 85.

```
> db.students.find({marks: {$gt: 85}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2223'),
    roll: 8,
    name: 'Pratik',
    age: 19,
    marks: 90,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
```

7. LESS THAN — $lt MongoDB

Query:

db. students. find ({marks: {$lt: 60}})

Explanation: Finds students whose marks are less than 60.



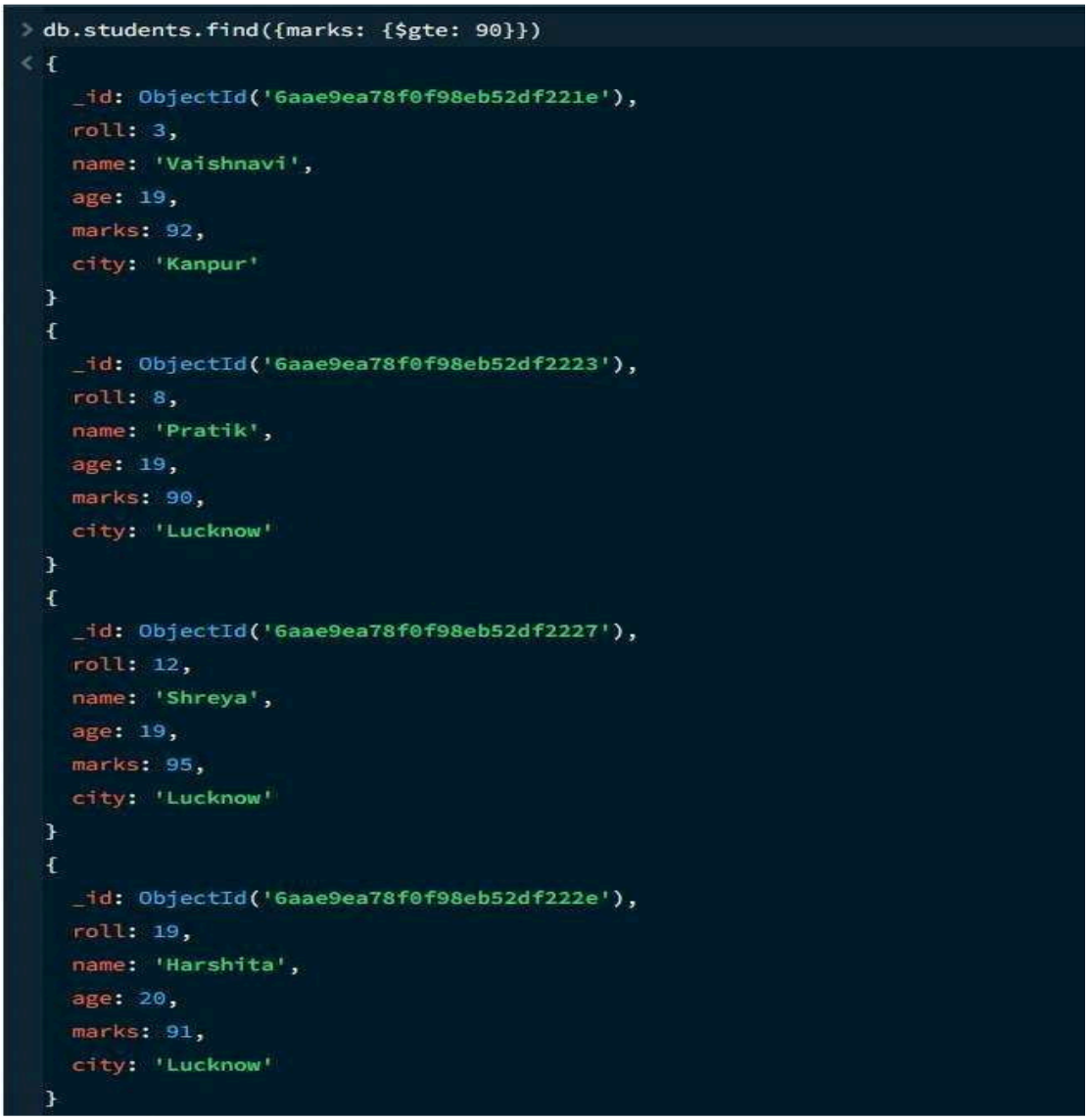
```
> db.students.find({marks: {$lt: 60}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df222d'),
    roll: 18,
    name: 'Jai',
    age: 18,
    marks: 58,
    city: 'Gonda'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df223c'),
    roll: 33,
    name: 'Jai',
    age: 20,
    marks: 55,
    city: 'Ayodhya'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df224a'),
    roll: 47,
    name: 'Aman',
    age: 18,
    marks: 59,
    city: 'Gonda'
  }
```

8. GREATER THAN OR EQUAL TO — $gte

MongoDB Query:

db. students. find({marks: {$gte: 90}})

Explanation: Finds students whose marks are 90 or above.



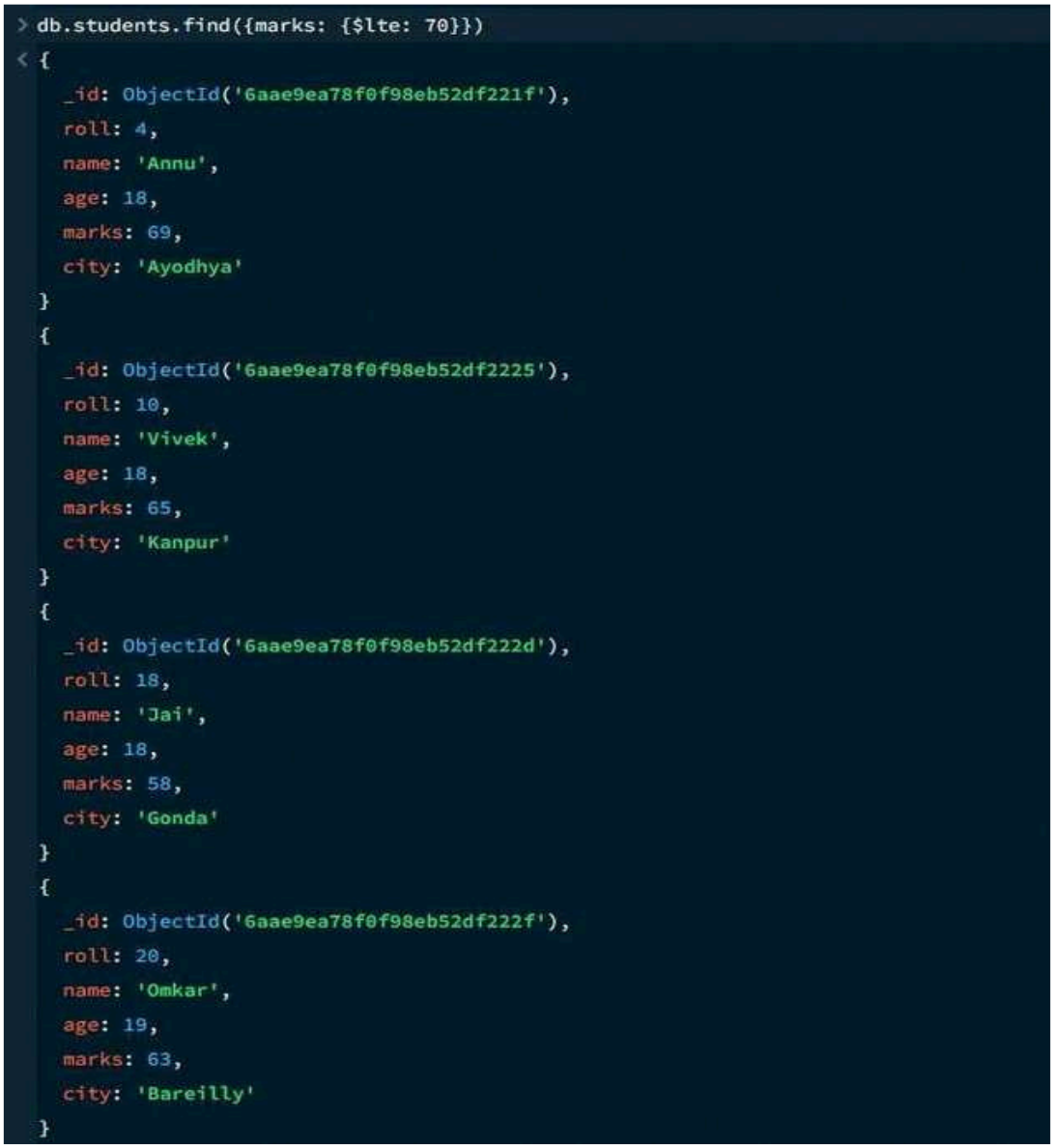
```
> db.students.find({marks: {$gte: 90}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2223'),
    roll: 8,
    name: 'Pratik',
    age: 19,
    marks: 90,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222e'),
    roll: 19,
    name: 'Harshita',
    age: 20,
    marks: 91,
    city: 'Lucknow'
  }
```

9. LESS THAN OR EQUAL TO

— $lte MongoDBQuery:

db.students.find({marks:{$lte:70}})

Explanation: Finds students whose marks are 70 or below.



10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(

  {roll:5},
  {$set:{marks:95}}
)
```

Explanation: Updates the marks to 95 of the student with roll number 5.

```
> db.students.updateOne(
    {roll:5},
    {$set:{marks:95}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.students.find({roll:5})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df2220'),
    roll: 5,
    name: 'Vansh',
    age: 19,
    marks: 95,
    city: 'Varanasi'
```

11. DELETE ONE OPERATION MongoDB

Query:

db.students.deleteOne({roll:10})

Explanation: Deletes one document matching roll number 10.

```
  }
> db.students.deleteOne({roll:10})
< {
    acknowledged: true,
    deletedCount: 1
  }
> db.students.find({roll:10})
<
Students >
```

12. DELETE MANY

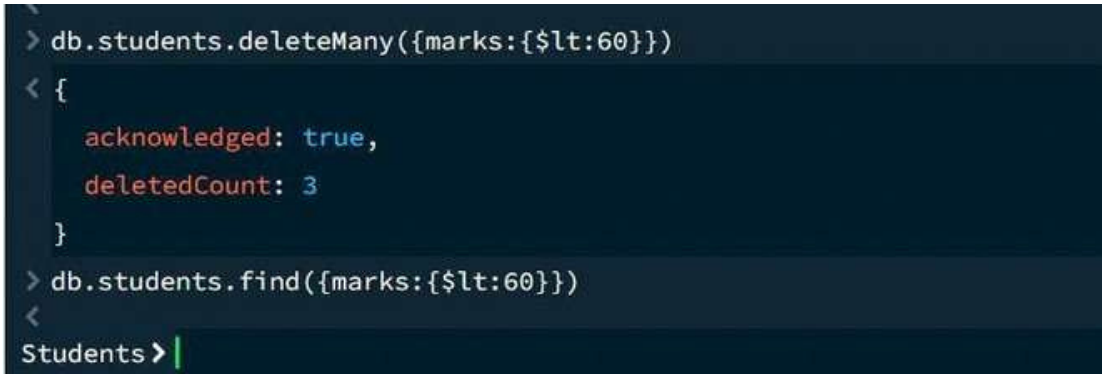
OPERATION MongoDB

Query:

```
db.students.deleteMany({

  marks: {$lt:60}
})
```

Explanation: Deletes all students having marks below 60.



13. AND OPERATION —

$and MongoDB Query:

```
db. students.find({

  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions.

```
>_MONGOSH
> db.students.find({
    $and:[
      {marks:{$gt:80}},{age:19}
    ]
  })
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
```

14. OR OPERATION — $or

MongoDB Query:

```
db.students.find({

  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})
```

Explanation: Finds students from Lucknow or Kanpur.

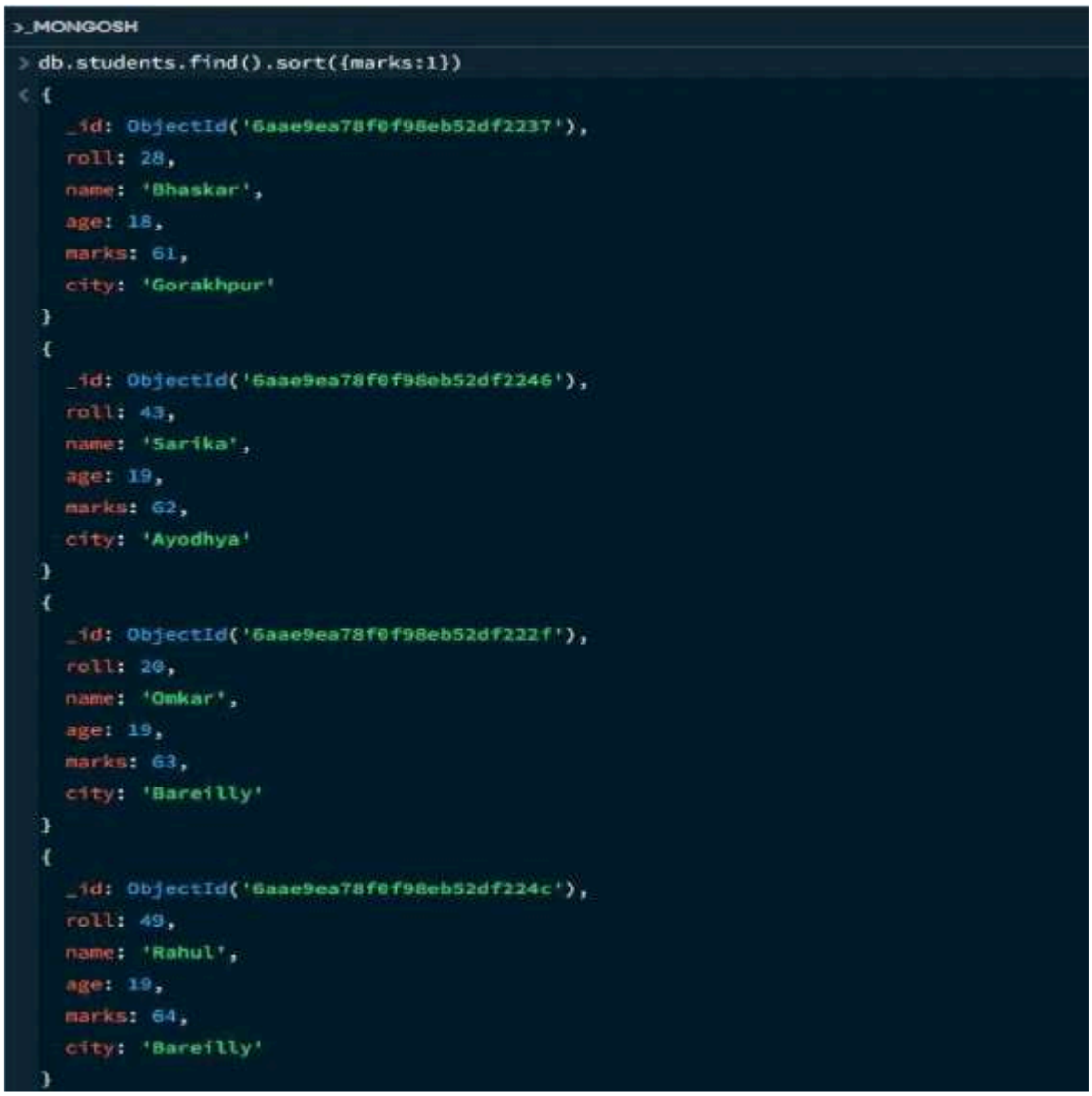
```
>_MONGOSH
> db.students.find({
    $or:[
      {city:"Lucknow"},{city:"Kanpur"}
    ]
  })
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
```

15. SORT OPERATION

A.MongoDB Query:

db. students. find (). sort({marks : 1})

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.



B. MongoDB Query:

db. students. find (). Sort ({marks: -1})

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
>_MONGOSH
> db.students.find().sort({marks:-1})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df223e'),
    roll: 35,
    name: 'Krishna',
    age: 19,
    marks: 96,
    city: 'Prayagraj'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2220'),
    roll: 5,
    name: 'Vansh',
    age: 19,
    marks: 95,
    city: 'Varanasi'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2235'),
    roll: 26,
    name: 'Anushka',
    age: 20,
    marks: 94,
    city: 'Lucknow'
  }
```

16. LIMIT OPERATION

MongoDB Query:

db. students. find (). limit (5)

Explanation: Displays only the first five documents.

```
>_MONGOSH
> db.students.find().limit(5)
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221d'),
    roll: 2,
    name: 'Uday',
    age: 18,
    marks: 76,
    city: 'Jaunpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
```

17. SEARCH BY CITY

MongoDBQuery:

db.students.find({city:"Lucknow"})

Explanation: Finds all students whose city is Lucknow.

```
>_MONGOSH
> db.students.find({city:"Lucknow"})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2221'),
    roll: 6,
    name: 'Amrita',
    age: 20,
    marks: 85,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2223'),
    roll: 8,
    name: 'Pratik',
    age: 19,
```

18. COUNT DOCUMENTS

MongoDB Query:

db. students. count Documents ()

Explanation: Counts documents currently present in the collection (performing after the delete function).

```
>_MONGOSH
  }
> db.students.countDocuments()
< 46
```

MONGODB OPERATORS USED:

Operator	Meaning	Example
$eq	Equalto	marks: {$eq:88}
$gt	Greater than	marks: {$gt:85}
$lt	Less than	marks: {$lt:60}
$gte	Greater than or equal to	marks: {$gte:90}
$lte	Less than or equal to	marks: {$lte:70}
$and	All conditions true	$and:[...]
$or	At least one condition true	$or:[...]

CONCLUSION

Thisprojectprovided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to Ascending order & descending order data, comparison operators such as $eq, $gt, $lt, $gte, and $lte work, along with logical operators such as $and and $or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.

MONGODB CRUD OPERATIONS

1. CREATE operation:

```
db.students.insertMany([
 { student_id: 1, name:
   "Aarav Sharma", age: 19,
   gender: "Male", course:
   "BCA Data Science",
   semester: 3, city:
   "Lucknow", marks: 82
},
{ student_id: 2, name:
   "Aditi Verma", age: 19,
   gender: "Female", course:
   "BCA Data Science",
   semester: 3, city:
   "Kanpur", marks: 88
},
{ student_id: 3, name:
"Aditya Singh", age:
20, gender: "Male",
course: "BCA AI",
semester: 3, city:
"Delhi", marks: 76
},
{  student_id:  4,  name:
age:  "Ananya  Gupta",
  19,
  gender:   "Female", course:
  "BCA AI",
  semester: 3, city:
  "Jaipur", marks:
  91
},
{ student_id: 5, name:
  "Arjun  Mehta",  age:  20,
  gender:  "Male",  course:
  "BCA Data Science",
```

```
  semester: 3, city: "Agra",
  marks: 74
},
{ student_id: 6, name:
  "Bhavya  Kapoor",  age:
  19,  gender:  "Female",
  course:    "BCA    AI",
  semester: 3, city:
  "Delhi", marks: 86

},
{ student_id: 7, name:
  "Chirag Yadav", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 79
},
{ student_id: 8, name:
  "Diya Mishra", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3, city:
     "Varanasi",
  marks: 93
},
{ student_id: 9, name: "Dev
Rajput", age: 20, gender:
"Male", course: "BCA Data
Science", semester: 3,

  city: "Meerut",
  marks: 81
},
```

```
{  student_id:  10,  name:
  "Ishita  Agarwal",  age:
  19,
  gender: "Female", course:
  "BCA AI",
  semester: 3,
  city: "Agra",
  marks: 89
},
{ student_id: 11, name:
  "Kabir Khan", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 72
},
{  student_id:  12,  name:
age:Kavya  Sharma",
  19,
  gender:   "Female", course:
  "BCA AI",
  semester: 3, city:
  "Jaipur", marks:
  95
},
{
  student_id: 13, name:
  "Manav Jain", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Kanpur", marks: 78
},
{ student_id: 14, name:
  "Mehak Sinha", age:
```

```
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3, city:
  "Delhi", marks:
  84
},
{ student_id: 15, name:
  "Naman  Gupta",  age:  20,
  gender:   "Male",   course:
  "BCA    Data    Science",
  semester: 3, city: "Agra",
  marks: 69
},
{ student_id: 16,
  name:        "Navya
  Singh",   age:   19,
  gender:   "Female",
  course:  "BCA  AI",
  semester:  3,  city:
  "Lucknow",   marks:
  90
},
{ student_id: 17, name: "Om
  Prakash", age: 20, gender:
  "Male", course: "BCA Data
  Science", semester: 3,
  city: "Varanasi", marks:
  75
},
{ student_id: 18, name:
  "Palak Verma", age:
  19, gender:
     "Female", course:
  "BCA AI",
```

```
  semester: 3,
  city:       "Kanpur",
  marks: 87
},
{ student_id: 19, name:
  "Raghav Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Jaipur", marks: 80
},
{
  student_id:      20,
  name: "Riya Gupta",
  age: 19, gender:
     "Female", course:
  "BCA AI", semester:
  3,
  city:       "Delhi",
  marks: 92
},
{ student_id: 21, name:
  "Rohan  Singh",  age:  20,
  gender:  "Male",  course:
  "BCA    Data    Science",
  semester:    3,    city:
  "Lucknow", marks: 77
},
{  student_id:  22,  name:
age:  Sakshi  Yadav",
  19,
  gender:   "Female", course:
  "BCA AI",
```

```
  semester: 3,
  city: "Agra",
  marks: 85
},
{ student_id: 23,
  name: "Samar Khan",
  age: 20, gender:
  "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Meerut", marks: 73


},
{ student_id: 24, name:
  "Simran Kaur", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3, city:
  "Delhi", marks:
  94
},
{ student_id: 25, name:
  "Tanishq Jain", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Kanpur", marks: 71
},
{ student_id: 26, name:
  "Tanvi Mishra",
  19,
  gender: "Female", course:
  "BCA AI",
age:
```

```
  semester: 3,
  city: "Varanasi",
  marks: 89
},
{ student_id: 27, name:
  "Uday Singh", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 83
},
{  student_id:  28,  name:
  "Vanshika  Gupta",  age:
  19,
  gender: "Female", course:
  "BCA AI",
  semester: 3,
  city: "Jaipur",
  marks: 96
},
{ student_id: 29, name:
  "Varun Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city: "Agra",
  marks: 68
},
{
  student_id:
     30, name: "Zoya
  Khan",   age:   19,
  gender:   "Female",
  course:  "BCA  AI",
  semester: 3,
  city:      "Delhi",
```

```
  marks: 91
},
{ student_id: 31, name:
  "Abhishek Verma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Kanpur", marks: 79
},
{  student_id:  32,  name:
age:Anjali  Sharma",
  19,
  gender:   "Female", course:
  "BCA      AI",
  semester: 3, city:
  "Lucknow", marks:
  88
},
{ student_id: 33,
name: "Deepak
Yadav", age: 20,
gender: "Male",

  course: "BCA Data
  Science", semester: 3,
  city: "Varanasi", marks:
  74
},
{  student_id:  34,  name:
age:Garima  Singh",
  19,
  gender:   "Female", course:
  "BCA AI",
```

```
  semester: 3,
  city: "Agra",
  marks: 90
},
{ student_id: 35, name:
  "Harsh Gupta", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Jaipur", marks: 82
},
{
  student_id: 36,
  name: "Isha Verma",
  age: 19, gender:
     "Female", course:
  "BCA AI", semester:
  3,
  city: "Delhi",
  marks: 93
},
{ student_id: 37, name:
  "Karan Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 76
},
{ student_id: 38, name:
age: "Lavanya Jain",
  19,
  gender: "Female", course:
  "BCA AI",
  semester: 3, city:
  "Kanpur", marks:
```

```
  87
},
{ student_id: 39, name:
  "Mohit Singh", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Meerut", marks: 70
},
{ student_id: 40,
  name: "Neha Gupta",
  age: 19, gender:
  "Female", course:
  "BCA AI", semester:
  3, city: "Jaipur",
  marks:
  95
},
{ student_id: 41, name:
  "Pranav Kumar", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city: "Agra",
  marks: 81
},
{ student_id: 42, name:
  "Priya Singh", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3,
  city:      "Lucknow",
  marks: 89
},
```

```
{ student_id: 43, name:
  "Rahul Verma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3,
  city: "Delhi",
  marks: 73
},
{  student_id:  44,  name:
"Ritika  Sharma",
age:
  19,
  gender:   "Female", course:
  "BCA      AI",
  semester: 3,
  city: "Kanpur",
  marks: 92
},
{ student_id: 45, name:
  "Shivam Gupta", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Varanasi", marks: 78
},
{ student_id: 46, name:
  "Shreya Jain", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3,
  city: "Agra",
  marks: 86
},
{
```

```
  student_id: 47, name:
  "Varsha Singh", age: 19,
  gender: "Female", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 84
},
{ student_id: 48,
  name: "Vivek
  Yadav", age: 20,
  gender: "Male",
  course: "BCA AI",
  semester: 3, city:
  "Jaipur", marks: 80
},
{ student_id: 49, name:
  "Yash Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Delhi", marks: 77
},
{ student_id: 50,
name: "Muskan
Verma", age: 19,
gender: "Female",

      name: 'Manav Jain', age:
       20, gender: 'Male',
       course: 'BCA Data
       Science', semester: 3,
       city: 'Kanpur', marks: 78
     }

     {
       _id:
       ObjectId('6aad46d1505f94ec0f315254'),
       student_id: 14, name: 'Mehak Sinha', age:
       19, gender: 'Female', course: 'BCA AI',
       semester: 3, city: 'Delhi', marks: 84
     }
```

```
{
  _id:
  ObjectId('6aad46d1505f94ec0f315255'),
  student_id: 15, name: 'Naman Gupta', age:
  20, gender: 'Male', course: 'BCA Data
  Science', semester: 3, city: 'Agra',
  marks: 69
}

{
  _id:
  ObjectId('6aad46d1505f94ec0f315256'),
  student_id: 16, name: 'Navya Singh', age:
  19, gender: 'Female', course: 'BCA AI',
  semester: 3, city: 'Lucknow', marks: 90
}

{
  _id:
  ObjectId('6aad46d1505f94ec0f315257'),
  student_id: 17, name: 'Om Prakash', age:
  20, gender: 'Male', course: 'BCA Data
  Science', semester: 3, city: 'Varanasi',
  marks: 75
}

{
  _id:
  ObjectId('6aad46d1505f94ec0f315258'),
  student_id: 18, name: 'Palak Verma', age:
  19, gender: 'Female', course: 'BCA AI',
  semester: 3, city: 'Kanpur', marks: 87
}

{
  _id:
  ObjectId('6aad46d1505f94ec0f315259'),
  student_id: 19, name: 'Raghav Sharma',
  age: 20, gender: 'Male', course: 'BCA
  Data Science', semester: 3, city:
  'Jaipur', marks: 80
}

{
  _id:   ObjectId('6aad46d1505f94ec0f31525a'),
   student_id: 20, name: 'Riya Gupta', age: 19,
  gender:   'Female',   course:   'BCA   AI',
  semester: 3, city: 'Delhi', marks: 92
}
```

```
db.students.findOne({student_id:10})
```

```
{
```

```
  _id:   ObjectId('6aad46d1505f94ec0f315250'),
  student_id:  10,  name:  'Ishita  Agarwal',
  age:  19,  gender:  'Female',  course:  'BCA
  AI', semester: 3,
  city: 'Agra',
  marks: 89
}
```

3. UPDATE operation:

```
db.students.updateOne({student_id:1},{$set:{marks:90}})
{ acknowledged:
  true, insertedId:
  null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

db.students.updateOne({students_id:5},{$set:{city:"Lucknow"}})
 { acknowledged:
  true, insertedId:
  null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
 }
db.students.findOne({student_id:5})

{
 _id:
  ObjectId('6aad46d1505f94ec0f31524b'),
  student_id: 5, name: 'Arjun Mehta', age:
  20, gender: 'Male', course: 'BCA Data
  Science', semester: 3, city: 'Agra',
```

```
  marks: 74
}
```

4. DELETE operation:

```
db.students.deleteOne({student_id: 40})

{ acknowledged:
true, deletedCount:
1 }



db.students.deleteMany({marks: {$lt: 70}})
{ acknowledged:
  true,
  deletedCount: 2
}
```